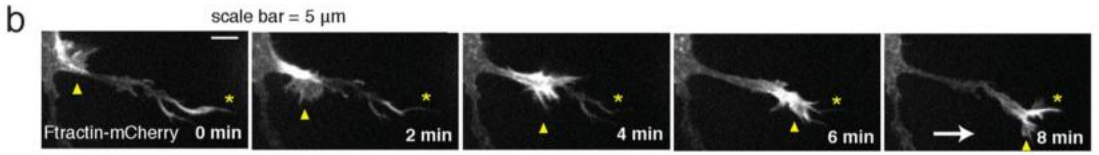
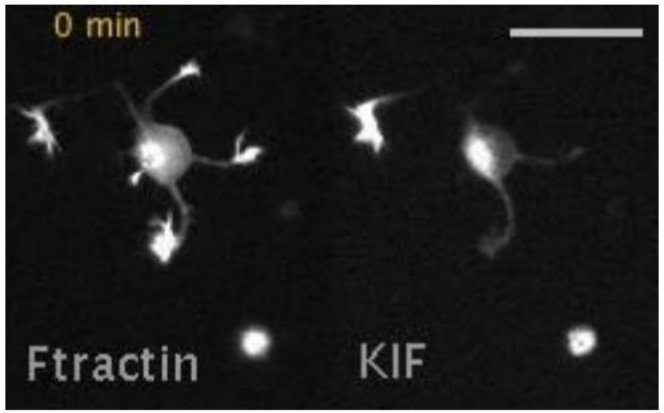
Actin waves accompany bursts of neurite growth

- **Actin waves** are traveling regions of **filamentous actin** that move outward through immature **neurites**.
  - A **neurite** is a growing projection that may become an axon or dendrite.
  - Researchers tracked fluorescent F-actin waves in cultured **rat hippocampal neurons**.
- A passing wave briefly **widens the neurite shaft** and rearranges its **cytoskeleton**.
  - **Microtubules** can extend into the wider region and provide tracks for cargo.
  - **Kinesin** moves cellular cargo along these microtubules toward the neurite tip.
- **Neurite extension occurs in bursts** when actin waves reach growing tips.
  - A neurite may remain stable and then lengthen rapidly as a wave arrives.
  - Waves reach different branches at different times, producing uneven growth.

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Winans et al. (2016), Fig. 1b. Developing neurites, actin, and intracellular transport.



Winans et al. (2016), Video 5. Academic: Neurite growth and transport.

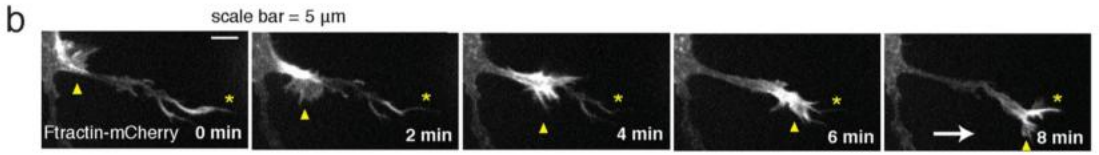
Actin waves accompany bursts of neurite growth

Developing neurons extend processes called **neurites**, which may later become axons or dendrites. **Actin waves** are moving regions rich in **filamentous actin**. Fluorescent labeling allowed Winans and colleagues to track these waves in cultured rat hippocampal neurons.

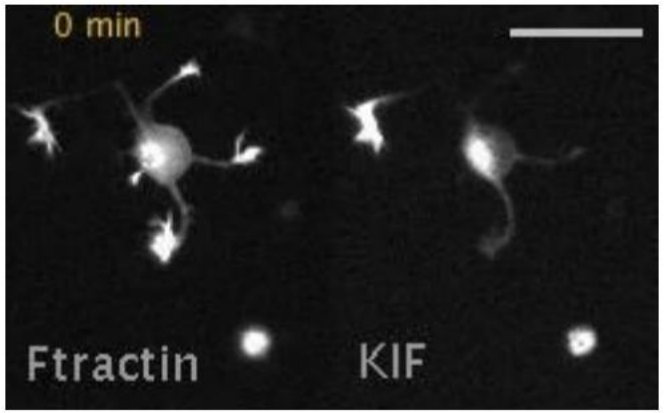
A passing actin wave briefly **widens the neurite shaft** and changes the local **cytoskeleton**. **Microtubules** extend into the wider region, providing structural support and tracks for intracellular cargo. **Kinesin motor proteins** move that cargo along microtubules toward the neurite tip.

Neurites do not always grow steadily. **Extension occurs in bursts** associated with arriving actin waves. One process may remain stable before lengthening rapidly, while another grows at a different time. Waves reaching individual branches at different times produce different patterns of neurite growth.

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Winans et al. (2016), Fig. 1b. Developing neurites, actin, and intracellular transport.



Winans et al. (2016), Video 5. Academic: Neurite growth and transport.